

OK FOOTWEAR LIMITED

Enterprise Resource Planning System

Technical Implementation Plan

Phase-wise Sprint Breakdown | Backend · Frontend · Test Cases

Version 1.0 | May 2025

Document	OK Footwear ERP — Technical Implementation Plan
Version	1.0
Phases	3 Phases 20 Sprints 12 Months
Sprint Length	2 weeks each
Methodology	Agile Scrum
Status	Draft — For Team Review
Classification	Confidential — Internal Use Only

Implementation Overview

The OK Footwear ERP is delivered in 3 phases over 12 months using 2-week sprints. Every sprint contains detailed backend tasks, frontend tasks, and test cases mapped to the Testing Strategy. No sprint closes without its associated tests passing in CI at the coverage gate (≥80% overall).

Col	Col	Col	Col	Col	Col
Phase 1	S1–S8	Months 1–4	Infrastructure, Auth, Orders, Procurement, Inventory, Finance Core	~110 BE ~85 FE	85+ cases
Phase 2	S9–S14	Months 5–8	Manufacturing (BOM/Prod/QC), HR, Payroll, ESS/MSS Portal	~95 BE ~80 FE	65+ cases
Phase 3	S15–S20	Months 9–12	Import/Export, Fixed Assets, Board Governance, Compliance, Analytics	~80 BE ~65 FE	32+ cases
Total	20	12 months	All 12 modules + infrastructure	~285 BE ~230 FE	182+ cases

PHASE 1 — Core Operations (Months 1–4)

Phase 1: Core Operations

Phase 1 establishes the project infrastructure and delivers the core operational cycle: authentication & RBAC, order management, procurement, inventory, and finance GL core. At the end of Phase 1 the system can process the full procure-to-dispatch workflow.

Sprint 1 Weeks 1–2 Infrastructure Setup, DevOps & Project Scaffolding

Backend Tasks

Task / Item	Details / Deliverable	Est.
Initialize NestJS project	TypeScript strict, Express adapter, path aliases (@/modules, @/common, @/shared), module structure	4h
Prisma v5 + multi-schema	schema.prisma previewFeatures=["multiSchema"]; 8 schemas created; PgBouncer transaction mode 20 conn	3h
PostgreSQL 16 baseline migration	Create all 8 schemas via prisma migrate; extension: uuid-ossdp, pg_trgm, pgcrypto, btree_gin	2h
Redis 7 configuration	ioredis client; DB0=queues, DB1=auth, DB2=cache; health check on startup	2h
BullMQ queue module	5 named queues: payroll, pdf, email, sms, report; dead-letter after 3 retries; Bull Board admin UI	3h
nestjs-pino logger	Correlation-id (UUID v7) injected via AsyncLocalStorage; X-Correlation-ID response header; JSON format	3h
@nestjs/config + Joi	Namespaced configs: DatabaseConfig, RedisConfig, AuthConfig, AwsConfig, SmsConfig; fail-fast on startup	2h
HttpExceptionFilter	RFC 7807 problem+json: type, title, status, detail, instance, correlationId, errors[]	3h
ResponseInterceptor	Wrap all 2xx responses in { data, timestamp } envelope; ClassSerializerInterceptor chain	1h
CORS + Helmet + Throttler	CORS whitelist from env; Helmet security headers; Redis sliding-window ThrottlerGuard	2h
Docker multi-stage Dockerfile	node:20-alpine builder + runner; non-root user; .dockerignore; target image < 200MB	3h
docker-compose.yml	PG16 + Redis7 + MinIO + PgBouncer + NestJS + Nginx; single command local startup	3h
GitHub Actions CI/CD pipeline	lint → type-check → unit → integration (testcontainers) → docker build → ECR push → k8s deploy	4h
Jest + testcontainers setup	ts-jest; testcontainers PG+Redis; coverage thresholds (80/75/80) enforced; transaction rollback pattern	3h
sys.document_sequences DDL	Table + sys.next_doc_number() function with row-level lock; seed: order, po, grn, payroll prefixes	2h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
React 18 + Vite 5 scaffold	TypeScript strict, ESLint + Prettier, path aliases (@/components, @/hooks, @/lib, @/stores)	3h

Task / Item	Details / Deliverable	Est.
Tailwind CSS + shadcn/ui	tailwind.config.ts; install shadcn CLI; seed: Button, Input, Dialog, Sheet, Badge, Tabs, Toast	2h
React Router v6	createBrowserRouter; lazy imports per route; RoleGuard HOC; 403/404 pages; nested layouts	3h
Axios API client	Base URL from VITE_API_URL; request interceptor (JWT); 401 refresh + retry; typed DTOs; toast on error	3h
TanStack Query v5	QueryClient (staleTime 30s, retry 1); ReactQueryDevtools in dev; default error handler to toast	2h
Zustand stores	authStore (userId, permissions[], accessToken); uiStore (locale, theme, sidebarOpen); notifStore (SSE)	3h
i18next EN/BN setup	/locales/en.json + bn.json; lazy-loaded; useTranslation; LanguageSwitcher; date/number formatting	2h
MSW + Vitest + RTL	MSW handlers directory; vitest.config.ts; setupTests.ts; jsdom; coverage thresholds; test utilities	3h
Playwright setup	playwright.config.ts; Chromium + Firefox; staging baseUrl; loginAs() + seedOrder() helpers	3h
AppShell (Layout)	Collapsible sidebar; topbar with user menu; notification bell; breadcrumb; responsive mobile drawer	4h
RoleGuard HOC	Reads authStore.permissions; can(module,action); redirects to /403; wraps all protected pages	2h
DataTable base component	TanStack Table v8; sortable columns; pagination; global search; column visibility toggle; export	4h
useAuth hook	can(module,action) from permissions[]; logout(); currentUser; permission-aware render helper	2h

Test Cases — Sprint 1

Task / Item	Details / Deliverable	Est.
TC-DB-SYS-001	Database	sys.next_doc_number() returns formatted number (ORD-000001 format)
TC-DB-SYS-002	Database	Counter increments by 1 on each invocation
TC-DB-SYS-003	Database	Concurrent calls (20 parallel) produce no duplicate numbers
TC-SEC-SESS-002	Security	All API responses include HSTS, CSP, X-Frame-Options, X-Content-Type-Options headers
TC-SEC-SESS-003	Security	CORS rejects requests from unknown origins (no ACAO header set)
TC-SEC-SESS-004	Security	CORS allows requests from whitelisted application origin
TC-SEC-AUTH-008	Security	Refresh token cookie has HttpOnly, Secure, SameSite=Strict flags

Sprint 2

Weeks 3–4

System Module — Auth, RBAC, Users, Audit, Notifications

Backend Tasks

Task / Item	Details / Deliverable	Est.
Schema: sys tables DDL	users (with failed_attempts, locked_until), roles, permissions, user_roles, role_permissions, refresh_tokens	3h
Schema: audit_logs partition	Yearly date-range partition; GIN index on new_value JSONB; composite index (table_name, record_id)	2h
Schema: notifications partition	Yearly partition; partial index on (user_id, is_read) for fast unread badge count query	2h
Schema: compliance_items	expiry_date, responsible_user_id, alert_days, status; partial index on status=valid	1h
AuthService.login()	argon2.verify; failed_attempts increment; logout at 5 (locked_until = now + 30min); generateTokens()	4h
AuthService.generateTokens()	accessToken (8h JWT HS256); refreshToken (30d); store refresh token hash to DB	2h
AuthService.refresh()	Validate token; check not in Redis blacklist; rotate (blacklist old token TTL=remaining life)	3h
AuthService.getPermissions()	DB query user→roles→permissions; JSON.stringify; Redis setex 300s; key: permissions:{userId}	2h
TotpService	authenticator.generateSecret(); AES-256-GCM encrypt secret; verify(token, decryptedSecret)	3h
AuthController	POST /auth/login, /logout, /refresh, /2fa/setup, /2fa/verify, /2fa/disable	2h
UserService + Controller	CRUD; role assignment endpoint; GET /users/me; soft-delete; employee account link	3h
RolesService + Controller	CRUD roles + permission matrix; invalidate Redis permission cache on change	2h
AuditInterceptor	Intercept all mutating requests (POST/PATCH/PUT/DELETE); record to audit_logs with old+new JSONB	4h
NotificationsService + SSE	EventSource /notifications/stream per user; BullMQ email+SMS dispatch; unread count endpoint	4h
ComplianceService + cron	Nightly 02:00: query items expiring within alert_days; send email to responsible_user	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Login page (complete)	React Hook Form + Zod; password visibility toggle; server error→field mapping; redirect after login	4h
2FA verification page	6-digit OTP input with auto-submit on 6th digit; countdown timer; resend code link	3h
Users management page	DataTable; Create/Edit user dialog; role assignment multi-select dropdown; status badge	4h
Roles & permissions page	Role list; permission matrix CheckboxGrid (module × action); optimistic save	5h
Audit log viewer page	DataTable with date/module/user filters; row expand shows old↔new JSON diff viewer	4h
Notification centre dropdown	Bell icon + badge; SSE connected indicator; mark-all-read; list of notifications with time	4h

Task / Item	Details / Deliverable	Est.
Compliance register page	DataTable; expiry traffic-light (green/amber/red by days remaining); countdown badge	3h
Profile page	Change password form; 2FA enable/QR code display; active sessions list with revoke button	4h
useNotifications hook	SSE EventSource; auto-reconnect on close; parse event data; update notifStore	3h
PermissionMatrix component	Checkbox grid controlled; select-all per row; disabled for system-protected permissions	4h

Test Cases — Sprint 2

Task / Item	Details / Deliverable	Est.
TC-AUTH-U-001	Unit	generateTokens() returns accessToken (8h) and sets httpOnly refresh cookie
TC-AUTH-U-002	Unit	Login throws UnauthorizedException on incorrect password
TC-AUTH-U-003	Unit	failed_attempts incremented; locked_until set when count reaches 5
TC-AUTH-U-004	Unit	Locked account rejected even with correct password
TC-AUTH-U-005	Unit	TOTP verification passes with valid OTP code
TC-AUTH-U-006	Unit	TOTP verification fails with expired code
TC-AUTH-U-007	Unit	Permission cache miss triggers DB lookup + Redis caching (300s TTL)
TC-AUTH-U-008	Unit	Refresh token rotation blacklists old token in Redis
TC-GUARD-U-001	Unit	RbacGuard returns true when user has required module:action permission
TC-GUARD-U-002	Unit	RbacGuard throws ForbiddenException when permission absent
TC-GUARD-U-003	Unit	ValidationPipe strips unknown properties (whitelist: true)
TC-GUARD-U-004	Unit	ThrottlerGuard throws ThrottlerException after limit exceeded
TC-AUTH-I-001	Integration	POST /api/auth/login → 200 with accessToken + httpOnly refreshToken cookie
TC-AUTH-I-002	Integration	POST /api/auth/login → 401 with RFC 7807 body on wrong password
TC-AUTH-I-003	Integration	POST /api/auth/refresh → new accessToken + rotated cookie
TC-AUTH-I-004	Integration	POST /api/auth/logout → session cleared; old refresh token rejected on retry
TC-FE-H-003	Frontend	useAuth.can() returns true for matching permission in authStore
TC-FE-H-004	Frontend	useAuth.can() returns false for missing permission
TC-FE-H-005	Frontend	useAuth.logout() clears store and navigates to /login
TC-SEC-AUTH-001	Security	Tampered JWT signature → 401
TC-SEC-AUTH-002	Security	Expired access token → 401 with "Token expired" detail
TC-SEC-AUTH-003	Security	JWT signed with wrong secret → 401
TC-SEC-AUTH-005	Security	Account locked (429) after 5 consecutive failed logins
TC-SEC-AUTH-007	Security	Old refresh token rejected (401) after rotation
TC-SEC-AUTHZ-001	Security	employee_ess cannot access payroll endpoint → 403
TC-E2E-AUTH-001	E2E	Login with correct credentials → dashboard
TC-E2E-AUTH-002	E2E	Wrong password → error message, stay on /login
TC-E2E-AUTH-003	E2E	Account locked after 5 failed attempts
TC-E2E-AUTH-004	E2E	MD role prompted for TOTP after password entry

Sprint 3

Weeks 5–6

Orders Module — Core CRUD, Status Machine, Buyers, Articles**Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: ord.buyers + articles	buyers (name, currency, payment_terms, credit_limit, country); articles (code, description, size_system, category)	2h
Schema: ord.orders + order_lines	orders (status, sample_approved, total_quantity, delivery_date, currency); order_lines (size_label, quantity)	2h
Schema: ord.order_milestones	milestones (order_id, milestone_type enum, planned_date, actual_date); 6 types auto-generated on confirm	1h
BuyersService + Controller	CRUD; GIN trigram search; GET /buyers (dropdown); soft-delete; currency config	3h
ArticlesService + Controller	CRUD; category/season filter; size_system link; GIN trigram search	3h
OrdersService.create()	Validate: line qty sum = totalQuantity; call next_doc_number(); set status=draft; link buyer+article	3h
OrdersService state machine	STATUS_TRANSITIONS map; sample gate (confirmed→in_production); OrderConfirmedEvent on confirm	4h
OrdersService.generateMilestones()	6 types: material_booking, pp_sample, bulk_start, qc, packing, shipment; backward from delivery_date	3h
OrdersController	GET list (paginated+filtered), GET detail (with lines), POST, PATCH, PATCH status, DELETE cancel	2h
OrderConfirmedEvent	Typed class with orderId, deliveryDate, buyerId, confirmedBy; emit after successful confirmation	1h
CreateOrderDto + validation	orderLines sum = totalQuantity; unitPrice > 0; deliveryDate future; currency ISO4217	2h
OrderResponseDto	@Exclude/@Expose; embedded buyer + article + orderLines; status machine next-allowed-states	2h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Orders list page	DataTable: order#, buyer, article, qty, delivery, status badge; filter by status/buyer/date range	4h
Order detail page	Master-detail: header info + milestone timeline + size breakdown + action buttons by status	5h
Create order wizard	Step 1: buyer+article+dates; Step 2: SizeRunInputGrid; Step 3: summary+submit	6h
SizeRunInputGrid component	Controlled grid; size cols from article size_system; running total; keyboard Tab navigation	4h
OrderStatusBadge component	Status → label + Tailwind colour; used in list and detail pages; with tooltip	1h
OrderStatusActions component	Contextual buttons by current status; confirm modal for irreversible actions	3h
MilestoneTimeline component	Vertical timeline; planned vs actual dates; overdue = red; completed = green checkmark	3h
Buyers management page	DataTable; create/edit slide-over panel; currency selector	3h
Articles management page	DataTable; create/edit form; size system selector; category tree	3h
useOrders hook	TanStack Query: list, detail, create, transitionStatus — optimistic updates on status change	3h

Test Cases — Sprint 3

Task / Item	Details / Deliverable	Est.
TC-ORD-U-001	Unit	Order number auto-generated in ORD-NNNNNN format from document sequence
TC-ORD-U-002	Unit	Status machine: draft → confirmed is a valid transition
TC-ORD-U-003	Unit	Invalid transition draft → in_production rejected with message
TC-ORD-U-004	Unit	confirmed → in_production blocked when sample_approved=false
TC-ORD-U-005	Unit	OrderConfirmedEvent fired with correct orderId payload on confirmation
TC-ORD-U-006	Unit	6 milestone records auto-generated with correct planned_dates
TC-ORD-I-001	Integration	POST /api/orders → 201 with auto-generated order number and status=draft
TC-ORD-I-002	Integration	POST /api/orders → 422 when orderLines sum ≠ totalQuantity
TC-ORD-I-003	Integration	GET /api/orders → paginated list with data[] + meta (page, limit, totalCount)
TC-ORD-I-004	Integration	PATCH /api/orders/:id/status → 200 on valid draft→confirmed transition
TC-ORD-I-005	Integration	PATCH /api/orders/:id/status → 422 when sample not approved for in_production
TC-ORD-I-006	Integration	GET /api/orders/:id → 404 with RFC 7807 body for non-existent order
TC-ORD-I-007	Integration	employee_ess POST /api/orders → 403 Forbidden
TC-FE-C-001	Frontend	OrderStatusBadge: all 5 statuses render correct label and CSS class
TC-FE-C-008	Frontend	SizeRunInputGroup: running total computed from size inputs
TC-FE-C-009	Frontend	SizeRunInputGroup: non-numeric input rejected
TC-FE-Z-001	Frontend	createOrderSchema: valid payload accepted without errors
TC-FE-Z-002	Frontend	createOrderSchema: empty orderLines array rejected
TC-FE-Z-003	Frontend	createOrderSchema: unitPrice=0 rejected with descriptive message
TC-E2E-ORD-001	E2E	Order manager creates order with full size breakdown on factory tablet
TC-E2E-ORD-003	E2E	in_production button disabled when sample_approved=false

Sprint 4

Weeks 7–8

Orders — Quotations, Samples, Complaints & CAPA**Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: ord.quotations	quotation_number, status (draft/sent/won/lost), bom_version_id, win_probability, outcome_reason	2h
Schema: ord.samples	round_number, sample_type enum (PP/counter/size_set/TOP), dispatch_date, approval_status	1h
Schema: ord.complaints + capa	complaints (type, severity, root_cause); capa_actions (description, owner_id, due_date, status)	2h
QuotationsService	create; auto-populate cost from approved BOM; close (won/lost); conversion rate KPI	4h
SamplesService	multi-round CRUD; approveSample() → sets sampleApproved=true on parent order	3h
ComplaintsService + CapaActionsService	create complaint; link to order; create/update CAPA actions; auto-close when all CAPA done	4h
Quotation + Sample + Complaint controllers	CRUD + workflow endpoints; nested under /orders/:id/	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Quotations tab (order detail)	Quotation list; create drawer with cost-sheet auto-populate; win/loss close modal	4h
Sample tracking tab	Multi-round timeline; per-round status; approve button with confirmation dialog	3h
Complaints & CAPA tab	Complaint form; CAPA action items with owner picker + due-date; progress badges	4h
E2E order test helpers	Playwright: createAndConfirmOrder(), approveAllSamples(), completeProductionCycle()	2h

Test Cases — Sprint 4

Task / Item	Details / Deliverable	Est.
TC-ORD-U-007	Unit	Quotation marks as lost with outcome_reason correctly stored
TC-ORD-U-008	Unit	CAPA action with past due_date rejected with validation error
TC-E2E-ORD-002	E2E	Full lifecycle: draft→confirmed→in_production→packed→delivered
TC-SEC-INJ-005	Security	Unknown DTO properties stripped by ValidationPipe whitelist

Sprint 5

Weeks 9–10

Procurement Module — Vendors, POs, GRN, Vendor Invoices**Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: prc all tables	vendor_categories, vendors, purchase_orders, po_lines, goods_receipts, gr_lines, gr_line_photos, vendor_invoices	3h
VendorsService	CRUD; approved/blacklisted enforcement; performance_rating auto-avg from GRN quality scores	3h
PurchaseOrdersService	Multi-level approval by amount threshold (config-driven); total_amount calc from lines; status machine	4h
GoodsReceiptsService	Create against PO; per-line accepted/rejected qty; QC photos S3 upload; fire GrnApprovedEvent on approve	4h
GrnApprovedEvent	Payload: grnId, lines[{itemId, warehouseId, acceptedQty, unitCost}]; handled by InventoryModule	1h
VendorInvoicesService	Three-way match ($PO \geq GRN \geq invoice$); TDS auto-deduct; link to GL AP posting; partial payment tracking	4h
Procurement controllers (all)	All CRUD + workflow actions; nested routes for PO lines, GRN lines, invoice matching	3h
PO approval workflow	Config-driven thresholds: <50k line mgr, <500k manager, <5M finance, >5M MD; email notification per step	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Vendor list + detail page	DataTable; status badge (approved/pending/blacklisted); performance score gauge; cert documents tab	4h
PO creation wizard	Step 1: vendor + expected_date + currency; Step 2: line items (item search, qty, price); Step 3: review	5h
PO approval queue page	Pending POs list; amount + threshold badge; approve/reject with mandatory reason; email trail	3h
GRN entry form	PO line items pre-loaded; per-line accepted/rejected qty; QC note; receipt photo upload	4h
Vendor invoice matching page	Three-way match visual ($PO \text{ amount} / GRN \text{ qty} / invoice$); tolerance % gauge; approve button	4h

Test Cases — Sprint 5

Task / Item	Details / Deliverable	Est.
TC-PRC-U-001	Unit	PO total_amount = sum of (quantity × unit_price) across all lines
TC-PRC-U-002	Unit	PO creation rejected for vendor with status=blacklisted
TC-PRC-U-003	Unit	Vendor invoice rejected when amount exceeds three-way match tolerance
TC-PRC-U-004	Unit	GrnApprovedEvent fired with item details on GRN approval
TC-PRC-U-005	Unit	GRN: accepted + rejected qty cannot exceed received qty
TC-SEC-AUTHZ-004	Security	finance_manager cannot access board module endpoints (403)
TC-SEC-INJ-001	Security	SQL injection in search parameter: OR 1=1 returns empty result
TC-SEC-INJ-002	Security	Non-UUID path parameter returns 400 Bad Request

Sprint 6

Weeks 11–12

Inventory Module — Transactions, Balances, Counts

Backend Tasks

Task / Item	Details / Deliverable	Est.
Schema: inv all tables	stock_items, warehouses, stock_transactions (yearly partition), stock_balances, stock_counts, stock_count_lines	3h
Trigger: inv.update_stock_balance	AFTER INSERT on stock_transactions; upsert stock_balances (qty += direction×qty); recalc weighted avg cost	4h
CHECK: non-negative balance	CONSTRAINT chk_balance_non_negative CHECK (quantity >= 0) on inv.stock_balances	1h
GrnApprovedHandler	@OnEvent grn.approved; insert stock_transactions (direction=+1) for each GRN line; log txn_type=grn	3h
StockItemsService + Controller	CRUD; category filter; GIN trigram search; reorder level config; below-reorder flag	3h
StockTransactionsService	recordMovement() — INSERT only, never UPDATE/DELETE; emit StockBelowReorderEvent when needed	3h
StockCountsService	Create count sheet (snapshot system_qty); enter physical_qty per item; variance = physical – system; approval workflow	4h
inv.stock_summary mat. view	CREATE MATERIALIZED VIEW CONCURRENTLY; total_qty, avg_cost, total_value, below_reorder; refresh daily cron	2h
Stock controllers + DTOs	All endpoints; paginated transaction history; GET /stock-summary (from materialized view)	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Stock items list page	DataTable; below-reorder badge in amber; category filter; reorder level column; edit drawer	3h
Stock balance view	Per-warehouse balance table; movements history accordion; avg_cost and total_value columns	3h
Stock count sheet page	Per-item physical count input grid; variance column (physical–system); colour-coded by variance %	4h
Warehouse management page	CRUD; warehouse type (raw/FG/packing/accessories); location text; active/inactive status	2h
Inventory KPI widget	Below-reorder count, pending stock counts, total SKUs — dashboard KPI cards with links	3h

Test Cases — Sprint 6

Task / Item	Details / Deliverable	Est.
TC-DB-INV-001	Database	Trigger: GRN insert increments balance by received qty
TC-DB-INV-002	Database	Trigger: production issue decrements balance correctly
TC-DB-INV-003	Database	Multiple transactions accumulate balance correctly (100+50–30=120)
TC-DB-INV-004	Database	CHECK constraint raises error when stock would go negative
TC-DB-INV-005	Database	Weighted average cost recalculated correctly on receipt (two batches test)
TC-DB-INV-006	Database	Average cost unchanged on stock-out (only receipts change avg cost)
TC-DB-MV-002	Database	inv.stock_summary below_reorder=true when qty ≤ reorder_level after refresh
TC-INV-U-001	Unit	StockTransactionsService calls create() — never update() or delete()

Task / Item	Details / Deliverable	Est.
TC-INV-U-002	Unit	StockBelowReorderEvent fired when balance $\leq$ reorder_level
TC-INV-U-003	Unit	No StockBelowReorderEvent fired when balance is above threshold
TC-INV-U-004	Unit	Variance = physical_qty – system_qty computed correctly
TC-INV-I-001	Integration	POST transaction → balance updated via trigger (verified in response)
TC-INV-I-002	Integration	GET stock-summary → returns total_qty and below_reorder flag
TC-E2E-INV-001	E2E	Approved GRN immediately reflects in stock balance screen
TC-E2E-INV-002	E2E	Stock count variance: store officer submits → Finance Manager approves

Sprint 7

Weeks 13–14

Finance Core — Chart of Accounts, GL, Periods, AP/AR, Delivery**Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: fin core tables	chart_of_accounts (hierarchy), gl_periods, gl_entries, gl_entry_lines (yearly partition), bank_accounts	3h
Schema: fin AR/AP tables	buyer_invoices (AR), vendor_invoices link, delivery_challans, challan_lines	2h
Trigger: fin.check_period_open	BEFORE INSERT on gl_entry_lines; EXCEPTION if parent period status IN (closed, locked)	2h
FinanceService.postJournal()	Validate debit=credit; check period open; bulk INSERT gl_entry_lines; assign correlation to source document	4h
GLService	CRUD journals; trial balance query; account balance (date range) using CTEs; period open/close/lock workflow	4h
ChartOfAccountsService	CRUD with parent_account_id hierarchy; prevent delete if has transactions; account type enum	3h
BankAccountsService	CRUD; link to GL cash account; statement import CSV/OFX; reconciliation status	3h
BuyerInvoicesService	AR invoices; ageing buckets; collection recording; dispute workflow; link to delivery challan	3h
DeliveryChallansService	Create from order; POD recording (photo+date); trigger AR invoice generation on confirmed delivery	3h
PayrollDisbursedHandler	@OnEvent payroll.disbursed; call FinanceService.postJournal() for salary expense + net payable GL entries	2h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
GL journal entry page	Form: narration, period picker, debit/credit line items; real-time balance indicator (must = 0 to submit)	5h
GL viewer / journal list	DataTable; expand row to see lines; filter by period/account/type/amount range; export to Excel	4h
Trial balance page	Account hierarchy tree; debit/credit columns; opening+movement+closing; period selector; Excel export	4h
AR aging page	Buyer invoices grouped by bucket (0-30, 31-60, 61-90, 90+d); colour-coded; collection entry panel	4h
GL periods management	List; open/close/lock buttons; unlock requires MD permission; confirm irreversible lock modal	3h
Chart of accounts tree	Expandable hierarchy; inline add child account; edit dialog; transaction count badge (no-delete guard)	4h
Delivery challans page	Create from confirmed order; POD photo upload; mark-delivered button; AR invoice auto-created	3h

Test Cases — Sprint 7

Task / Item	Details / Deliverable	Est.
TC-FIN-U-001	Unit	GL journal rejected when total debit ≠ total credit
TC-FIN-U-002	Unit	Balanced journal (debit=credit) posts successfully
TC-FIN-U-003	Unit	Posting to locked period throws UnprocessableEntityException
TC-FIN-U-004	Unit	Straight-line depreciation: (cost–salvage) / usefulLife / 12

Task / Item	Details / Deliverable	Est.
TC-FIN-U-005	Unit	Diminishing balance: $\text{net_book_value} \times \text{annualRate} / 12$
TC-DB-FIN-001	Database	Trigger: posting to open period succeeds
TC-DB-FIN-002	Database	Trigger: posting to closed period raises exception
TC-DB-FIN-003	Database	Trigger: posting to locked period raises exception
TC-DB-CON-001	Database	GL line: debit and credit cannot both be non-zero (chk constraint)
TC-DB-CON-002	Database	GL line: debit and credit cannot both be zero (chk constraint)
TC-FIN-I-001	Integration	POST /api/gl/entries → 201 with status=posted for balanced journal
TC-FIN-I-002	Integration	POST /api/gl/entries → 422 for unbalanced journal with error detail
TC-FIN-I-003	Integration	POST /api/gl/entries → 422 when posting to locked period
TC-PERF-001	Performance	Orders list endpoint: 50 VUs × 3min → p95 < 300ms, error < 0.1%

Sprint 8 (Week 15–16): Phase 1 regression sweep, staging deployment, UAT preparation. All Phase 1 test cases re-executed in CI. k6 baseline benchmarks captured.

PHASE 2 — Production & HR (Months 5–8)

Phase 2: Production & HR

Phase 2 delivers manufacturing operations (BOM versioning, production scheduling, daily output tracking, QC), the full HR and payroll engine compliant with Bangladesh Labour Act 2006, and the ESS/MSS self-service portals.

Sprint 9 Weeks 1–2 (Ph2) Manufacturing — BOM Versioning, Cost Sheets

Backend Tasks

Task / Item	Details / Deliverable	Est.
Schema: mfg BOM tables	bom_headers (article_id, version, status, approved_by), bom_lines (item_id, qty_per_unit, wastage_pct), bom_size_overrides	3h
Schema: mfg.cost_sheets	material_cost, labour_cost, overhead_cost, target_margin_pct, selling_price (computed); links to bom_header	2h
BomService.create()	New version per article; status=draft on create; duplicate from previous version option; approval workflow	4h
BomService.approve()	Set status=approved; only one active version per article enforced; previous versions archived	3h
ProductionBlockGuard	Check approved BOM exists for article before allowing production order creation	2h
CostSheetsService	Auto-generate from BOM × latest vendor rates; material/labour/overhead split; selling price with margin	4h
BOM + CostSheet controllers	CRUD; GET /articles/:id/bom; GET /bom/:id/cost-sheet; POST /bom/:id/approve; version history list	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
BOM viewer page	Tree/table view of BOM lines; wastage % column; size overrides section; version history tabs; compare versions	5h
BOM create/edit form	Component line items with item search + qty + wastage_pct; size override per SKU; duplicate version button	4h
BOM approval UI	Submit-for-approval button; approver review with comment; approved/pending/rejected badge; audit trail	3h
Cost sheet page	Auto-calculated breakdown; margin % slider → selling price real-time update; print/Excel export	4h

Test Cases — Sprint 9

Task / Item	Details / Deliverable	Est.
TC-MFG-U-001	Unit	BOM ConflictException when same article+version already exists
TC-MFG-U-002	Unit	Production order blocked without approved BOM for article
TC-MFG-U-005	Unit	Cost sheet: $\text{selling_price} = \text{total_cost} \times (1 + \text{marginPct}/100)$
TC-DB-CON-003	Database	Order line quantity=0 rejected by positive CHECK constraint
TC-DB-CON-004	Database	Duplicate size_label in same order rejected by UNIQUE constraint

**Sprint
10–11**

Weeks 3–6 (Ph2)

**Manufacturing — Production Orders, Daily Entry, QC, Machines,
Scrap****Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: mfg production tables	production_orders, daily_productions (yearly partition), qc_results (defects JSONB), machines, machine_maintenance, scrap_records	3h
DailyScheduler: nightly lock	02:00 BDT cron: UPDATE daily_productions SET locked=true WHERE prod_date < today AND locked=false	2h
ProductionOrdersService	Create from sales order + BOM version; planned_qty per size; status machine; link to factory line	3h
DailyProductionService	Record per-operation per-shift output; efficiency_pct generated column; locked check; BullMQ nightly lock	3h
QcResultsService	AQL sampling size auto-calc; defect entry per type (JSONB); verdict pass/fail/rework; production order link	3h
MachineService	CRUD; maintenance log; downtime_hours generated column (sum maintenance durations); link to fixed asset	2h
ScrapService	Record by type + production order; disposal auth workflow; disposal method enum	2h
ProductionCompletedEvent	Fired when production order QC passes; handled by OrdersModule to update order status to qc	1h
Production controllers (all)	Full CRUD + workflow for production orders, daily entry, QC, machines, scrap	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Production schedule page	Calendar + list view; planned vs actual qty progress bars; status filters; drill into daily entries	4h
Daily production entry (tablet)	Touch-optimised; large tap targets; per-shift entry; running total vs target; locked-day indicator	5h
QC form page	AQL sampling: sample size auto-calculated; defect entry by type; pass/fail/rework verdict + notes	4h
QC analytics page	Defect frequency chart (Recharts bar); first-pass yield trend; top 5 defect types doughnut	4h
Machine register page	List; maintenance schedule; downtime log entry form; utilisation % KPI card; link to asset	3h
Production efficiency widget	Real-time efficiency % per factory line; target vs actual grouped bar chart; drill-down to daily	3h

Test Cases — Sprints 10–11

Task / Item	Details / Deliverable	Est.
TC-MFG-U-003	Unit	efficiency_pct = (produced / target) × 100 computed as generated column
TC-MFG-U-004	Unit	efficiency_pct is NULL when target_qty is 0
TC-MFG-U-006	Unit	Locked daily production entry throws exception on update attempt
TC-DB-PART-001	Database	2025-dated rows route to the 2025 partition child table
TC-DB-PART-002	Database	2025 rows excluded when querying with WHERE txn_date >= 2026-01-01

**Sprint
12–13**Weeks 7–10
(Ph2)**HR Module — Employees, Leave, Attendance, PF, Gratuity****Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: hr core tables	departments, designations, employees, employee_secrets (BYTEA for NID/passport/bank AES-256), employment_events	3h
Schema: hr leave tables	leave_types, leave_policies, leave_balances, leave_requests; partial index on status=pending	2h
Schema: hr attendance tables	attendance_records (yearly partition); biometric_sync_log; manual_corrections table	2h
Schema: hr PF + gratuity	pf_accounts, pf_transactions, gratuity_provisions; hr.compute_gratuity() function with Labour Act formula	3h
hr.compute_gratuity() function	Basic $\times (30/26) \times \text{years}$; ≥ 6 months rounds up to next year; < 1 year returns 0	3h
EmployeesService	Full lifecycle; AES-256-GCM encrypt NID/bank/passport at write; decrypt only when explicitly requested	4h
LeaveService	apply() with conflict detection (overlapping check); balance deduction; accrual engine; carry-forward	5h
AttendanceService	biometricSync() upsert; manualCorrection() with reason; calculateLop(month) for payroll use	4h
PfService	Monthly contribution calculation (10%+10%); annual interest credit cron; statement generation	3h
GratuityService	Monthly provision via compute_gratuity(); accrue to gratuity_provisions; GL expense posting via FinanceService	3h
HR controllers/DTOs (all)	All endpoints for employees, leave, attendance, PF, gratuity	4h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Employee directory page	DataTable with avatar, dept, designation, join date, status; click → detail page	4h
Employee detail/edit page	Tabs: personal, employment, salary, leave, attendance, documents; edit each section in slide-over	5h
Leave management page (HR)	Balance overview by type; leave request list; approval queue; team leave calendar heatmap	4h
Attendance dashboard	Monthly grid heatmap; P/A/L/LOP per cell; LOP days count; manual correction form	4h
Leave type config page	CRUD leave types; policy settings (accrual rate, max days, carry-forward limit, encashment)	3h
Org chart + departments page	OrgChartTree component; CRUD departments; designation management; headcount KPI	3h
PF account statement page	Per-employee: monthly contributions, interest, balance; CSV export; employer contribution breakdown	3h

Test Cases — Sprints 12–13

Task / Item	Details / Deliverable	Est.
TC-HR-U-001	Unit	LOP deduction proportional: $\text{basic}/\text{workingDays} \times \text{lopDays}$ (parametric test 3 cases)

Task / Item	Details / Deliverable	Est.
TC-HR-U-002	Unit	PF deduction at 10% of basic salary
TC-HR-U-005	Unit	Gratuity formula: 6yr, 5y6m, 5y5m, <1yr — all correct
TC-HR-U-006	Unit	Leave application rejected when available balance < requested days
TC-HR-U-007	Unit	Half-day leave deducts 0.5 days from balance
TC-HR-U-008	Unit	Overlapping leave request detected and rejected
TC-HR-U-009	Unit	Salary advance instalment = amount / recoveryMonths
TC-DB-HR-001	Database	hr.compute_gratuity(): 6 years → correct BDT amount to 2dp
TC-DB-HR-002	Database	hr.compute_gratuity(): 5y 6m rounds up to 6 years
TC-DB-HR-003	Database	hr.compute_gratuity(): 5y 5m stays at 5 years
TC-DB-HR-004	Database	hr.compute_gratuity(): < 1 year returns 0
TC-DB-CON-005	Database	factory_category required for factory employee type (chk constraint)
TC-FE-Z-004	Frontend	leaveRequestSchema: endDate before startDate rejected with message
TC-SEC-ENC-001	Security	NID stored as BYTEA (not plain text) in employee_secrets
TC-SEC-ENC-002	Security	Bank account field unreadable from DB without decryption key

Sprint 14

 Weeks 11–14
(Ph2)

Payroll Engine, Bank File Export, ESS/MSS Portal
Backend Tasks

Task / Item	Details / Deliverable	Est.
Schema: hr payroll tables	payroll_runs (status machine), payroll_entries (per employee), salary_structures, salary_components	2h
PayrollService.initiateRun()	Validate no duplicate, prior GL period closed, all attendance locked; enqueue BullMQ payroll job	3h
PayrollProcessor (BullMQ)	Per employee: basic+allowances–LOP–PF_employee–TDS+OT+festival_bonus; 30-min timeout; progress %	6h
PayrollService GL posting	On Finance Manager approval: FinanceService.postJournal() for salary expense + net payable	3h
Bank disbursement file export	BRAC Bank / Dutch-Bangla / IBBL format; configurable column mapping; CSV/fixed-width download	4h
PDF payslip BullMQ job	Handlebars template → Puppeteer render → S3 stream → return presigned URL (15min TTL)	4h
ESS scoped endpoints	/ess/leave, /ess/payslip, /ess/expense, /ess/attendance, /ess/advance — all scoped to req.user.employeeId	4h
MSS approval queue endpoints	Scoped to manager direct reports; leave/attendance/expense/overtime; SLA overdue flag; bulk approve	4h
ExpensesService + SalaryAdvancesService	Expense claim with receipt S3 upload; advance request with recovery schedule; approval workflow	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Payroll run wizard	4-step: period selection → pre-run checks → computation (polling) → approve + bank file download	6h
Payroll run detail page	Employee-wise entries; LOP/OT/PF/TDS breakdown columns; expand per-employee detail; approve button	4h
ESS dashboard	Leave balance cards; upcoming holidays; payslip download shortcut; expense pending count	3h
ESS leave application page	Type selector; date range picker with calendar; half-day toggle; balance indicator; history list	4h
ESS payslip page	Monthly list; Download PDF button using usePayslipDownload hook; loading spinner; toast on error	3h
ESS expense claim page	Claim form with receipt photo upload; item description; claim history with status badge	3h
ESS attendance self-view	Monthly calendar; P/A/L/LOP per day; correction request form with reason	3h
MSS approval queue page	Items by type (leave/expense/OT); unread badge; SLA overdue highlight; bulk select+approve	5h
MSS team calendar page	Monthly calendar showing all team members leave; conflict detection overlay with warning	4h

Test Cases — Sprint 14

Task / Item	Details / Deliverable	Est.
TC-HR-U-003	Unit	Festival bonus: included in Eid month, zero in non-qualifying months

Task / Item	Details / Deliverable	Est.
TC-HR-I-001	Integration	POST /api/payroll/runs → 202 Accepted, queues BullMQ job
TC-HR-I-002	Integration	POST /api/payroll/runs → 409 when run already exists for period
TC-HR-I-003	Integration	POST /api/leave-requests → 201 with status=pending
TC-HR-I-004	Integration	PATCH /api/leave-requests/:id/approve → 200 with status=manager_approved
TC-HR-I-005	Integration	Employee cannot GET another employee payslips → 403
TC-FE-C-006	Frontend	Payroll wizard: step 1 requires period selection before proceeding to step 2
TC-FE-C-007	Frontend	Processing step polls every 5 seconds until status=Processed
TC-FE-H-001	Frontend	usePayslipDownload: success fetches URL and triggers browser download
TC-FE-H-002	Frontend	usePayslipDownload: API error → status=error + destructive toast
TC-SEC-AUTHZ-002	Security	Employee IDOR: cannot access another employee payslip by ID substitution
TC-SEC-AUTHZ-003	Security	Manager cannot view employees from another department (403)
TC-E2E-PAY-001	E2E	Full payroll run: HR initiates → Finance approves → employee downloads payslip PDF
TC-E2E-PAY-002	E2E	Duplicate run for same month shows conflict error
TC-E2E-LEAVE-001	E2E	Leave apply → manager approve → balance visibly decremented
TC-E2E-LEAVE-002	E2E	Insufficient balance blocks leave submission at form level
TC-CONT-HR-001	Contract	GET /api/employees/:id — Pact response shape verified on both consumer+provider
TC-CONT-HR-002	Contract	GET payslip presigned URL — Pact shape matches consumer expectation
TC-PERF-006	Performance	Concurrent payroll report downloads: 20 VUs × 2min, all complete within 30s

PHASE 3 — Compliance & Analytics (Months 9–12)

Phase 3: Compliance & Analytics

Phase 3 completes the full ERP: import/export LC lifecycle, fixed assets with depreciation, budget management, board governance (Companies Act 1994 full compliance), the compliance register, and the analytics dashboard.

Sprint 15–16

Weeks 1–4 (Ph3)

Import/Export LCs, Fixed Assets, Depreciation, Budgets

Backend Tasks

Task / Item	Details / Deliverable	Est.
Schema: fin.import_lcs + export_lcs	LC lifecycle tables; document_submission tracking; tolerance_pct; payment_terms; landed_cost_lines	3h
Schema: fin.fixed_assets	category_id, purchase_date, original_cost, salvage_value, useful_life_years, depreciation_method, net_book_value (generated)	2h
Schema: fin.budgets + budget_lines	Annual budget header (year, type); monthly line items per account; version control	2h
ImportLcService	Full lifecycle: open → utilise → customs → landed cost → settlement; landed cost allocation by weight/value	4h
ExportLcService	Open → negotiate → present docs → bank acceptance → payment → BB repatriation; tolerance check	4h
FixedAssetsService	CRUD; SL/DB depreciation computation; disposal with gain/loss; link to GL fixed asset account	4h
MonthlyScheduler: depreciation	1st month cron: compute monthly depreciation per active asset; postJournal() for depreciation expense	3h
BudgetService	Annual OPEX/CAPEX budgets; monthly lines; budget vs actual variance endpoint (from GL actuals)	4h
LC + Asset + Budget controllers	All CRUD + workflow endpoints; document upload to S3; depreciation schedule report endpoint	3h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Import LC lifecycle page	Stepper: open→utilise→customs→landed cost→settlement; document checklist; landed cost table	5h
Export LC management page	Document submission tracker; tolerance gauge; repatriation status; overdue highlight	4h
Fixed assets register page	DataTable: code, category, NBV, depreciation method; depreciation schedule modal; disposal form	4h
Budget vs actual page	Monthly comparison table; variance column (favourable=green, adverse=red); drill-down to GL	4h
Finance dashboard (complete)	GL period status, AP/AR aging charts, fixed asset NBV KPI, budget variance summary cards	4h

Test Cases — Sprints 15–16

Task / Item	Details / Deliverable	Est.
TC-FIN-U-004	Unit	SL depreciation: $(originalCost - salvageValue) / usefulLifeYears / 12$

Task / Item	Details / Deliverable	Est.
TC-FIN-U-005	Unit	DB depreciation: $\text{netBookValue} \times \text{annualRate} / 12$
TC-FIN-U-006	Unit	Budget variance: negative (adverse) when actual > budget
TC-PERF-005	Performance	GL account balance query: 30 VUs × 3min → p95 < 400ms

**Sprint
17–18**

Weeks 5–8 (Ph3)

**Board Governance — Directors, Shares, Meetings, Resolutions,
AGM, Dividends****Backend Tasks**

Task / Item	Details / Deliverable	Est.
Schema: brd all tables	directors, shareholders, share_transactions (append-only), board_meetings, meeting_agenda, meeting_attendees, resolutions, agms, dividends, dividend_payments, related_parties	4h
brd.current_shareholding mat. view	Aggregate share_transactions; pct_held computed; CONCURRENTLY refresh; partial index for fast lookups	2h
DirectorsService	CRUD; RJSC DIN; AES-256-GCM encrypt NID/passport; appointment/resignation with Companies Act tracking	3h
ShareholdersService + share register	CRUD; share certificate issuance; append-only share_transactions (allotment/transfer/buyback)	3h
BoardMeetingsService	Create meeting; quorum_required config; attendee check-in; quorum validation blocks minutes finalisation	4h
ResolutionsService	Draft → pass; SHA-256 hash of signed PDF; DocuSign envelope send; webhook callback handler	4h
AgmService	15-month interval enforcement; overdue alert; notice dispatch tracking (statutory 14 days); proxy register	3h
DividendsService	Declare per_share amount; auto-compute per shareholder from mat. view; WHT deduction per NBR rate	3h
ResolutionSignedEvent handler	Verify DocuSign HMAC-SHA256 webhook; download signed PDF; store S3; record sha256_hash in DB	3h
RelatedPartiesService	IAS 24 register; transaction alert when other modules involve related parties; disclosure report	2h
Board controllers/DTOs (all)	All endpoints for directors, shareholders, meetings, resolutions, AGM, dividends, related parties	4h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Director register page	DataTable; appointment/resignation history; encrypted fields visible to Company Secretary only	4h
Shareholder register page	Register of Members; transfer history; certificate issuance form; current % holdings from mat. view	4h
Shareholding chart	Doughnut chart of current share % from materialized view; click slice → shareholder detail	3h
Board meetings page	Meeting creation; agenda builder; attendee check-in (QR or manual); quorum indicator badge	5h
Meeting minutes + resolutions	Rich text editor for minutes; resolution drafting form; pass/defer/withdraw actions	4h
Resolution detail with hash	Signed PDF viewer; SHA-256 hash displayed; DocuSign signing status badge; tamper indicator	3h
AGM management page	AGM schedule list; days-since-last-AGM counter (red if >14 months); notice dispatch checklist	3h
Dividend declaration page	Per-share amount input; shareholder breakdown table with WHT column; gross/net split; approve button	4h
Related parties register page	IAS 24 register; recent transaction alert badge; disclosure summary report export	3h

Test Cases — Sprints 17–18

Task / Item	Details / Deliverable	Est.
TC-BRD-U-001	Unit	Board meeting minutes blocked when attendee count < quorum_required
TC-BRD-U-002	Unit	SHA-256 hash stored correctly on resolution signing
TC-BRD-U-003	Unit	Dividend WHT deducted correctly at configured % (parametric: 10%, 15%)
TC-BRD-U-004	Unit	AGM flagged overdue when > 15 months since last AGM date
TC-DB-MV-001	Database	brd.current_shareholding reflects allotments after REFRESH MATERIALIZED VIEW
TC-SEC-SESS-006	Security	Audit log written for every employee update (AuditInterceptor test)
TC-E2E-BRD-001	E2E	Full board resolution: schedule meeting → pass resolution → Chairman eSign → hash recorded

**Sprint
19–20**Weeks 9–12
(Ph3)**Compliance, Analytics Dashboard, Full Regression & Handover****Backend Tasks**

Task / Item	Details / Deliverable	Est.
ComplianceService (complete)	Licence/cert register; nightly expiry check cron; email notification N days before expiry	3h
Analytics KPI endpoints	Order fill rate, production efficiency, stock turnover, payroll cost trend, AR collection rate	4h
Report generation jobs (all)	ExcelJS .xlsx: payroll summary, trial balance, inventory valuation, HR headcount, compliance status	4h
Full Swagger documentation review	All endpoints annotated; OpenAPI 3.0 spec exported; request/response examples complete	3h
Performance profiling + N+1 fixes	Prisma query logging; identify slow queries; add missing indexes; optimize heavy CTE queries	4h
Grafana dashboards finalization	All custom dashboards set up; alert thresholds configured; PagerDuty/Slack webhook tests	3h
Mutation testing (Stryker)	Run Stryker on all service files; identify weak assertions; fix test gaps; report to team	5h

Frontend Tasks

Task / Item	Details / Deliverable	Est.
Analytics dashboard (main)	KPI grid: open orders, production efficiency, stock alerts, payroll cost, AR overdue, GL status, compliance	6h
Order analytics page	Fill rate trend chart; buyer-wise analysis table; on-time delivery %; complaint rate trend	4h
Compliance register page (complete)	Expiry traffic lights; days-remaining countdown; responsible user assignment; bulk export	3h
Report centre page	Scheduled + on-demand reports list; download button (presigned URL); schedule email delivery option	3h
Chromatic visual regression finalization	All Storybook stories complete; push to Chromatic; approve all visual baselines	3h
Accessibility audit (WCAG 2.1 AA)	Run axe-core on all pages; fix violations; keyboard navigation; screen reader labels	3h
Full Bangla translation review	Native speaker review all bn.json strings; date/number formatting; placeholder translations	2h

Test Cases — Sprints 19–20 (Full Regression)

Task / Item	Details / Deliverable	Est.
TC-PERF-001	Performance	Orders list: 50 VUs × 3min → p95 < 300ms, error rate < 0.1%
TC-PERF-002	Performance	Dashboard KPI: all 4 endpoints under 200ms at normal load
TC-PERF-003	Performance	Employee search: 100 VUs × 2min → p95 < 200ms
TC-PERF-004	Performance	Stock summary: 200 VU ramp-up → p95 < 500ms sustained
TC-PERF-007	Performance	Spike test: 500 VU login burst → p99 < 2s, error rate < 2%
TC-PERF-008	Performance	Biometric sync: 10 devices × 5min → p95 < 200ms per punch
TC-PERF-009	Performance	30-minute soak test: 50 VUs → no memory leak, p95 < 300ms throughout

Task / Item	Details / Deliverable	Est.
TC-SEC-INJ-003	Security	XSS payload in remarks field not reflected in API response
TC-SEC-INJ-004	Security	Oversized 11MB payload rejected with 413 response
TC-SEC-SESS-001	Security	Rate limiting: 429 returned after >100 requests/minute per user
TC-SEC-SESS-005	Security	passwordHash and totpSecret never appear in any API response
TC-CONT-ORD-001	Contract	GET /api/orders/:id — Pact consumer+provider contract verified
TC-CONT-ORD-002	Contract	GET /api/orders paginated list shape verified
TC-CONT-ORD-003	Contract	POST /api/orders 422 RFC 7807 error shape verified
TC-CONT-AUTH-001	Contract	POST /api/auth/login success shape verified
TC-CONT-AUTH-002	Contract	POST /api/auth/login 401 error shape verified
TC-CONT-INV-001	Contract	GET /api/inventory/stock-summary shape verified
TC-CONT-SSE-001	Contract	SSE notification event structure verified
TC-E2E-RBAC	E2E	All RBAC matrix entries verified: blocked roles cannot access restricted pages/APIs
Full regression suite	All types	All 182 test cases pass; overall coverage ≥ 80%; zero Sentry regressions on staging

Implementation Summary

The OK Footwear ERP implementation spans 20 sprints over 12 months across 3 phases. Every sprint is gated by the CI test suite — no sprint closes without its test cases passing at the $\geq 80\%$ coverage threshold.

Col	Col	Col	Col	Col	Col	Col
Phase 1 — Core Ops	S1–S8	Months 1–4	Infrastructure, Auth+RBAC, Orders, Procurement, Inventory, Finance Core	~110	~85	85+ cases
Phase 2 — Production+HR	S9–S14	Months 5–8	Manufacturing (BOM/Prod/QC), HR, Payroll, ESS/MSS	~95	~80	65+ cases
Phase 3 — Compliance	S15–S20	Months 9–12	LCs, Fixed Assets, Budgets, Board Governance, Analytics	~80	~65	32+ cases
TOTAL	20 sprints	12 months	All 12 modules + infrastructure	~285	~230	182+ cases

— End of Document —